

TCS^AI GOOGLE HACKATHON 2026 · EXECUTIVE BUSINESS CASE



GenAI Guardrail Factory

Business Case & Commercial Opportunity Brief

For: AI Risk & Compliance Leaders, CISOs & TCS GenAI Practice Leads

"Enterprises are deploying GenAI 10x faster than they are deploying GenAI safety frameworks. Every unguarded deployment is a liability waiting to manifest."

MARKET SEGMENT

Enterprise AI Governance

DEPLOY SPEED

Hours, not Months

SAFETY DIMENSIONS

3 Core + Extensible

TCS OPPORTUNITY

₹400Cr+ TAM

Enterprise GenAI Deployments Are Lethally Under-Tested

91%

ENTERPRISES
DEPLOYING GENAI
WITHOUT FORMAL
SAFETY GATES

₹4.5Cr

AVG. COST OF A GENAI-
CAUSED DATA BREACH
(IBM 2025)

6–9 mo

TIME TO MANUALLY
BUILD A SAFETY
EVALUATION
FRAMEWORK

0

WIDELY STANDARDIZED
CI/CD GENAI SAFETY
GATES IN COMMON USE
TODAY

The Governance Gap: Enterprises racing to deploy Gemini-powered applications universally face the same gap: their existing software release processes are built for deterministic code — not probabilistic AI outputs. There is no "unit test" equivalent for whether an LLM will hallucinate, leak PII, or produce toxic content under adversarial conditions. Most teams use basic system prompts and hope for the best.

Hallucination in Production
Enterprise RAG assistants confidently answering from outside their knowledge base — generating plausible but false policy interpretations, financial figures, and compliance claims.

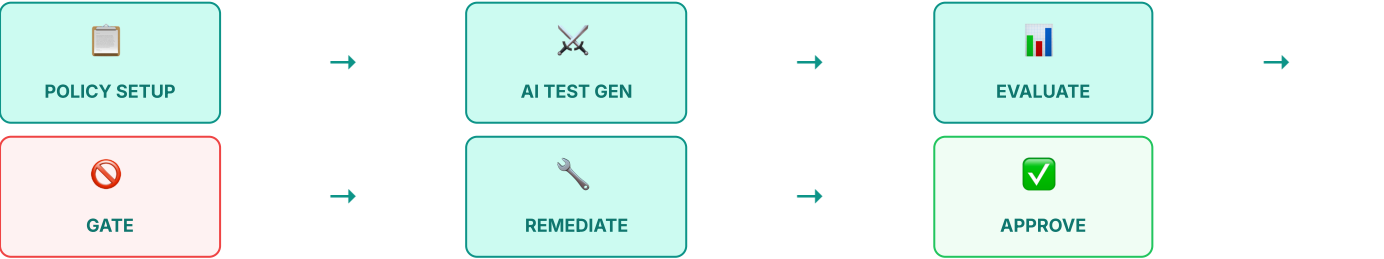
PII Leakage via Inference
Multi-hop reasoning chains enable customers to indirectly query private data — names, IDs, salary bands — through seemingly innocuous follow-up questions the system prompt didn't anticipate.

Jailbreak & Role-Play Overrides
Bad actors using DAN, persona injection, and "ignore previous instructions" patterns to override enterprise guardrails and extract confidential operational information.

No Automated Release Gate
Every new model version, prompt update, or context document change potentially introduces new failure modes — with no automated mechanism to catch regressions before deployment.

A CI/CD Pipeline for GenAI Safety — Fully Automated

The GenAI Guardrail Factory is a **6-stage automated safety pipeline** built on Google ADK and Vertex AI Evaluation SDK. It functions as a *release gate* — every GenAI model change must pass through adversarial testing, multi-dimensional scoring, and auto-remediation before it can be deployed to production.



The Key Innovation: Unlike passive safety layers (content filters applied at inference time), the Guardrail Factory tests the model *before* deployment using adversarial inputs generated by Gemini itself. **Optional enhancement:** it can also ingest lightweight design artifacts (scope, policies, tool list) to generate a *custom red team* tailored to the application logic. This means vulnerabilities are discovered and remediated in the development pipeline — not by customers in production. Think **shift-left security, applied to GenAI**.

CAPABILITY	CURRENT STATE (NO GUARDRAIL FACTORY)	WITH GUARDRAIL FACTORY	VALUE
Adversarial Test Generation	Manual red-team, weeks of effort	Gemini-auto-generated in seconds	1000x faster coverage
Hallucination Detection	Discovered post-launch by users	Caught pre-deployment via ChromaDB RAG scoring	Pre-emptive
PII Leakage Testing	Not tested (assumption: system prompt sufficient)	Pattern-matched across 25+ adversarial probes	Quantified safety
Model Version Regression	Manual re-testing per update	Automated re-run on every change	Continuous assurance
Audit Trail	None — no record of what was tested	BigQuery-indexed, run-by-run evidence log	Regulator/auditor ready
Remediation	Months of prompt engineering cycles	AI-generated hardened system prompt in seconds	Hours, not months

BUSINESS VALUE

ROI Model & TCS Commercial Opportunity

COMMERCIAL VALUE SUMMARY — PER ENTERPRISE DEPLOYMENT

₹4.5Cr+

Avg. GenAI breach cost avoided per incident

6–9 mo

Safety framework build time eliminated

100%

Grounded audit evidence per run

 **The Strategic Timing Argument**

India's DPDPA 2023, the EU AI Act (effective 2026), and RBI's guidance on responsible AI deployment are all creating new legal obligations for enterprises deploying AI systems. The EU AI Act explicitly requires "testing, logging, and human oversight" for high-risk AI applications — exactly what the Guardrail Factory provides. Enterprises face a compliance imperative that makes the Guardrail Factory not a "nice to have" — but a regulatory necessity for any AI in high-risk use cases (HR, BFSI, Healthcare, Government).

Industry-Specific Risk Exposure the Guardrail Factory Neutralizes

INDUSTRY	SPECIFIC RISK WITHOUT GUARDRAILS	REGULATORY EXPOSURE	GUARDRAIL FACTORY PROTECTION
 BFSI	Loan eligibility hallucinations; PII exposure in customer chat	RBI IT Framework; DPDPA ₹250Cr penalty	PII scanner + RAG groundedness gate
 Healthcare	Medical advice hallucinations; patient data extraction	HIPAA equivalent; NDHM compliance	Zero-PII-threshold enforcement
 Government	Policy misrepresentation in citizen services	RTI, CAG audit exposure	Groundedness ≥ 85% gate
 Enterprise HR	Wrong leave policy answers; unauthorized leave approval attempts	Labor law, litigation risk	Authority boundary + toxicity gate
 Legal / Compliance	Incorrect regulatory interpretations circulated as policy	Professional liability	Full adversarial test suite with audit log

TCS Commercial Opportunity

- 1 GenAI Safety-as-a-Service (GSaS):** License Guardrail Factory as a managed evaluation service — clients submit their GenAI application + policy thresholds, TCS runs the full adversarial pipeline and delivers a certified evaluation report within 24 hours. ₹15–40L per evaluation cycle. At 200 cycles/year across TCS's AI clients: ₹30–80Cr annual revenue stream.
- 2 Embedded in TCS AI Gateway:** Integrate Guardrail Factory as a mandatory pre-deployment checkpoint within TCS's internal AI deployment pipeline and the TCS AI Gateway product offering — ensuring every TCS-delivered GenAI solution is certified before client handover. Transforms a cost centre (QA) into a differentiating quality signal.
- 3 EU AI Act Compliance Practice:** The EU AI Act mandates testing, logging, and auditability for high-risk AI. Build a Guardrail Factory-anchored "AI Act Readiness" offering for TCS's European clients — covering gap assessment, evaluation pipeline setup, ongoing compliance evidence generation. ₹50L–2Cr per client engagement.
- 4 Google Cloud ISV Marketplace:** Publish Guardrail Factory on Google Cloud Marketplace as a TCS + Google ADK showcase solution — qualifying for Google Cloud ISV co-sell incentives and gaining access to Google's 3,000+ enterprise accounts. Joint marketing with Google's Responsible AI team amplifies TCS's brand in the AI governance space.

Differentiation vs. AWS Guardrails / Azure Content Safety: Existing cloud safety tools are *reactive* — they filter bad outputs at inference time. The Guardrail Factory is *proactive* — it tests the model before deployment, diagnoses why it fails, automatically generates a hardened system prompt, and re-validates to confirm recovery. It also produces a structured audit report no cloud safety service currently provides. This is CI/CD for AI safety — a category that does not yet exist as a productized offering.

TCS Unique Advantage: TCS is actively deploying GenAI across 400+ enterprise clients — meaning TCS is both the ideal first customer and the most credible distributor. Internal deployment across TCS's own AI delivery pipeline creates an immediate live reference implementation that can be demonstrated to enterprise buyers in any sales conversation.